

UNIT I

Matrices

The concept of matrices has had its origin in various types of linear problems, the most important of which concerns the nature of solutions of any given system of linear equations. Matrices are also useful in organizing and manipulating large amounts of data. Today, the subject of matrices is one of the most important and powerful tools in Mathematics which has found applications to a very large number of disciplines such as Engineering, Business and Economics, statistics etc.

1.1 Definition of a matrix

Definition 1.1.1: A rectangular arrangement of mn numbers (real or complex) in to m horizontal rows and n vertical columns enclosed by a pair of brackets $[]$, such as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m1} & \dots & a_{mn} \end{bmatrix}$$

is called an $m \times n$ (read “m by n”) **matrix** or a matrix of **order** $m \times n$.

Parentheses () are also commonly used to enclose numbers constituting matrices.

Before we go any further, we need to familiarize ourselves with some terms that are associated with matrices. The numbers in a matrix are called the **entries** or the **elements** of the matrix. For the entry a_{ij} , the first subscript i specify the row and the second subscript j the column in which the entry appears. That is, a_{ij} is an element of matrix A which is located in the i^{th} row and j^{th} column of the matrix A. Whenever we talk about a matrix, we need to know the **order** of the matrix.

The **order** of a matrix is the number of rows and columns it has. When we say a matrix is a 3 by 4 matrix, we are saying that it has 3 rows and 4 columns. The rows are always mentioned first and the columns second. This means that a 3×4 matrix does not have the same order as a 4×3 matrix. It must be noted that even though an $m \times n$ matrix contains mn elements, the entire matrix should be considered as a single entity. In keeping with this point of view, matrices are denoted by single capital letters such as **A**, **B**, **C** and so on.

Remark: By the size of a matrix or the dimension of a matrix we mean the order of the

matrix.

Example 1.1.1: Let $A = \begin{bmatrix} 1 & 5 & 2 \\ 0 & 3 & 6 \end{bmatrix}$.

Solution: Since A has 2 rows and 3 columns, we say A has order 2×3 , where the number of rows is specified first. The element 6 is in the position a_{23} (read a two three) because it is in row 2 and column 3.

Example 1.1.2: What is the value of a_{23} and a_{32} in $A = \begin{bmatrix} -1 & 4 & 7 \\ 2 & 3 & 1 \\ 5 & 7 & 8 \end{bmatrix}$?

Solution: a_{23} , the element in the second row and third column, is 1 and a_{32} , the element in the third row and second column, is 7. What is the size of this matrix?

Activity 1.1.1: 1. Suppose A is a 5×7 matrix, then

- a. A has 7 rows. (True/False)
- b. a_{ij} is an element of A for $i = 6$ and $j = 4$. (True/False)
- c. For what values of i and j , a_{ij} is an element of A?

2. Suppose $A = \begin{bmatrix} 4 & -7 & 5 \\ 8 & 1 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 8 \\ -7 & 1 \\ 5 & 6 \end{bmatrix}$

- a. What is the order of A and B?
- b. Find a_{22} , a_{13} , b_{13} and b_{31} .

It is customary to abbreviate the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

by the symbol $(a_{ij})_{m \times n}$ or more simply (a_{ij}) . This notation merely indicates what type of symbols we are using to denote the general entry.

Example 1.1.3: Form a 4 by 5 matrix, B , such that $b_{ij} = i + j$.

Solution: Since the number of rows is specified first, this matrix has four rows and five columns.

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} & b_{15} \\ b_{21} & b_{22} & b_{23} & b_{24} & b_{25} \\ b_{31} & b_{32} & b_{33} & b_{34} & b_{35} \\ b_{41} & b_{42} & b_{43} & b_{44} & b_{45} \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \end{bmatrix}.$$

Activity 1.1.2: Form a 4 by 3 matrix, B , such that

$$\text{a) } b_{ij} = i \times j \qquad \text{b) } b_{ij} = (-1)^{i+j}$$

Definition 1.1.2: Two matrices A and B are said to be **equal**, written $A = B$, if they are of the same order and if all corresponding entries are equal.

For example, $\begin{bmatrix} 5 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 2+3 & 1 & 0 \\ 2 & 3 & 2 \times 2 \end{bmatrix}$ but $\begin{bmatrix} 9 & 2 \end{bmatrix} \neq \begin{bmatrix} 9 \\ 2 \end{bmatrix}$. Why?

Example 1.1.4: Given the matrix equation $\begin{bmatrix} x+y & 6 \\ x-y & 8 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 3 & 8 \end{bmatrix}$. Find x and y .

Solution: By the definition of equality of matrices, $\begin{matrix} x+y=1 \\ x-y=3 \end{matrix}$, solving gives $x = 2$ and $y = -1$.

Activity 1.1.3: Find the values of x , y , z and w which satisfy the matrix equation

$$\begin{aligned} \text{a. } & \begin{bmatrix} x-y & 2x+z \\ 2x-y & 3z+w \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix} \\ \text{b. } & \begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2w \end{bmatrix} \end{aligned}$$

1.2. Types of matrices: Square, identity, scalar, diagonal, triangular, symmetric, and skew symmetric matrices

We now turn our attention to certain special types of matrices, which play important roles in matrix theory

Row Matrix: A matrix of order $1 \times n$ (one row, n columns) is called a **row matrix** or **row vector**

For example, the matrix $A = \begin{bmatrix} 5 & 2 & -1 & 4 \end{bmatrix}$ is a row matrix of order 1×4 .

Column Matrix: A matrix of order $m \times 1$ (m rows, one column) is called a **column matrix** or **column vector**.

For example, the matrix $B = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ is a 3×1 column matrix.

Zero or Null Matrix: A matrix whose entries are all 0 is called a **zero** or **null matrix**. It is usually denoted by $0_{m \times n}$ or more simply by **0**. For example, $0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is a 2×4 zero matrix.

Square Matrix: An $m \times n$ matrix is said to be a **square matrix** of **order n** if $m = n$. That is, if it has the same number of **columns** as **rows**.

For example, $\begin{bmatrix} -3 & 4 & 6 \\ 2 & 1 & 3 \\ 5 & 2 & -1 \end{bmatrix}$ and $\begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix}$ are square matrices of order 3 and 2 respectively.

➤ In a square matrix $A = (a_{ij})$ of order n , the entries $a_{11}, a_{22}, \dots, a_{nn}$ lying on the diagonal that extends from the **upper left corner** to the **lower right corner** are called the **main diagonal entries**, or simply the **main diagonal**

Thus, in the matrix $C = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 6 & 0 \\ 5 & 1 & 8 \end{bmatrix}$ the entries $c_{11} = 3$, $c_{22} = 6$ and $c_{33} = 8$ constitute the main diagonal.

Note: The sum of the entries on the main diagonal of a square matrix A of order n is called the **trace of A** .

That is, $\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn} = \sum_{i=1}^n a_{ii}$

❖ The trace is only defined for **square matrices**.

Activity 1.2.1: Find the trace of C in the above example.

Triangular Matrix: A square matrix is called an **upper triangular** matrix if all the entries below its main diagonal are **zero**. Similarly, it is called a **lower triangular** matrix if all the entries above its main diagonal are **zero**.

For example, $\begin{bmatrix} 2 & 4 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{bmatrix}$ and $\begin{bmatrix} 5 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 6 & 1 & 2 & 0 \\ -2 & -4 & 8 & 6 \end{bmatrix}$ are upper and lower triangular matrices, respectively.

Diagonal Matrix: A square matrix is called a diagonal matrix if all entries outside the main diagonal are zero.

Thus a square matrix $A = (a_{ij})$ is diagonal if $a_{ij} = 0$ for $i \neq j$.

For example, $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ is a diagonal matrix.

Notation: A diagonal matrix A of order n with diagonal elements $a_{11}, a_{22}, \dots, a_{nn}$ is denoted by

$$A = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$$

Scalar matrix: A diagonal matrix in which all the diagonal entries are equal is called a **scalar** matrix

For example, $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is a scalar matrix.

Note: Let $A = (a_{ij})$ be a square matrix. A is a scalar matrix if and only if $a_{ij} = \begin{cases} k, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$.

Identity Matrix or Unit Matrix: A square matrix is called an **identity matrix (or unit matrix)** if all its main diagonal entries are equal to 1 and all other entries are 0. In other words, it is a diagonal matrix whose diagonal elements are all 1. An identity matrix of order n is denoted by I_n , or simply by I

For example, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is identity matrix of order 3. $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is identity matrix of order 2.

Note: Let $A = (a_{ij})$ be a square matrix. A is an identity matrix if and only if

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

- The identity matrix acts as the **multiplicative identity** in matrix multiplication:

That is, $AI = IA = A$ for any square matrix A of the same order.

- It is both a **diagonal matrix** and a special case of a **scalar matrix** (with scalar value 1).

Symmetric matrix: A square matrix is called a **symmetric matrix** if it is equal to its transpose. In other words, a matrix **A** is symmetric when $A = A^T$. This means that the entries are mirrored across the main diagonal: $a_{ij} = a_{ji}$ for all i, j .

For example matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 6 \\ 4 & 6 & 7 \end{bmatrix}$. Transpose of A is; $A^T = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 6 \\ 4 & 6 & 7 \end{bmatrix}$

Since $A = A^T$, the matrix A is symmetric

Skew symmetric matrices: A square matrix is called a **skew-symmetric matrix** if it is equal to the negative of its **transpose**. In other words, a matrix **A** is skew-symmetric when $A^T = -A$. This means that each entry satisfies $a_{ij} = -a_{ji}$, and all diagonal entries must be zero.

For example matrix $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 4 \\ 3 & -4 & 0 \end{bmatrix}$. Transpose of A is; $A^T = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix}$

Clearly, $A^T = -A$. So, A is **skew-symmetric**.

1.3 Algebra of matrices

Activity 1.3.1: $\begin{bmatrix} 30 & 18 \\ 20 & 12 \\ 16 & 10 \end{bmatrix} + \begin{bmatrix} 36 & 20 \\ 24 & 18 \\ 20 & 12 \end{bmatrix} = \begin{bmatrix} 66 & ? \\ ? & 30 \\ ? & ? \end{bmatrix}$. Can you guess what number should appear in the entries

marked by question mark?

Addition of matrices: Let A and B be two matrices of the same order. Then the addition of A and B, denoted by $A + B$, is the matrix obtained by adding corresponding entries of A and B. Thus, if

$$A = (a_{ij})_{m \times n} \text{ and } B = (b_{ij})_{m \times n}, \text{ then } A + B = (a_{ij} + b_{ij})_{m \times n}.$$

Remark: Notice that we can add two matrices if and only if they are of the same order. If they are, we say they are conformable for addition. Also, the order of the sum of two matrices is same as that of the two original matrices.

Activity 1.3.2: Given the matrices A, B, and C below

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 1 \\ 5 & 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 & 3 \\ 2 & 4 & 2 \\ 3 & 6 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$$

Find, if possible. **a)** $A + B$ **b)** $B + C$

If A is any matrix, the negative of A , denoted by $-A$, is the matrix obtained by replacing each entry in A by its negative. For example, if

$$A = \begin{bmatrix} 2 & -1 \\ 5 & 4 \\ -6 & 0 \end{bmatrix}, \text{ then } -A = \begin{bmatrix} -2 & 1 \\ -5 & -4 \\ 6 & 0 \end{bmatrix}$$

Properties of Addition of Matrices

1. Matrix addition is commutative. That is, if A and B are two matrices of the same order, then $A + B = B + A$.
2. Matrix addition is associative. That is, if A , B and C are three matrices of the same order, then $(A + B) + C = A + (B + C)$.
3. Existence of additive identity. That is, if 0 is the zero matrix of the same order as that of the matrix A , then $A + 0 = A = 0 + A$.
4. Existence of additive inverse. That is, if A is any matrix, then

$$A + (-A) = 0 = (-A) + A$$

Note: The zero matrix plays the same role in matrix addition as the number zero does in addition of numbers.

Subtraction of Matrices: Let A and B be two matrices of the same order. The difference $A - B$ is defined as $A + (-B)$. In other words, $A - B$ is obtained by subtracting each entry of B from the corresponding entry of A .

Example 1.3.1: Let $A = \begin{bmatrix} 4 & -1 \\ 2 & 3 \\ 5 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & -2 \\ 6 & 1 \end{bmatrix}$

$$\text{Then } \mathbf{A} - \mathbf{B} = \begin{bmatrix} 4-0 & -1-2 \\ 2-5 & 3-(-2) \\ 5-6 & -7-1 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -3 & 5 \\ -1 & -8 \end{bmatrix}$$

Scalar Multiplication of a Matrix

Let $\mathbf{A}$ be a matrix and k a real number (called a **scalar**). The scalar multiplication of $\mathbf{A}$ by k , denoted by $k\mathbf{A}$, is the matrix obtained by multiplying **each entry of $\mathbf{A}$ by k** .

This operation is called **scalar multiplication**.

Example 1.3.2: If $\mathbf{A} = \begin{bmatrix} 0 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 7 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix}$. Find $2\mathbf{A} + 3\mathbf{B}$.

Solution: $2\mathbf{A} = 2 \begin{bmatrix} 0 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 6 \\ 4 & 2 & 8 \end{bmatrix}$ and $3\mathbf{B} = 3 \begin{bmatrix} 7 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 21 & 18 & 9 \\ 3 & 12 & 15 \end{bmatrix}$

$$2\mathbf{A} + 3\mathbf{B} = \begin{bmatrix} 0 & 4 & 6 \\ 4 & 2 & 8 \end{bmatrix} + \begin{bmatrix} 21 & 18 & 9 \\ 3 & 12 & 15 \end{bmatrix} = \begin{bmatrix} 21 & 22 & 15 \\ 7 & 14 & 23 \end{bmatrix}$$

Example 1.3.3: Express the matrix equation $x \begin{bmatrix} 2 \\ 1 \end{bmatrix} - y \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 2 \begin{bmatrix} 8 \\ 11 \end{bmatrix}$ as a system of equations and solve.

Solution: The given matrix equation gives

$$\begin{bmatrix} 2x \\ x \end{bmatrix} - \begin{bmatrix} 3y \\ 5y \end{bmatrix} = \begin{bmatrix} 16 \\ 22 \end{bmatrix} \Rightarrow \begin{bmatrix} 2x - 3y \\ x - 5y \end{bmatrix} = \begin{bmatrix} 16 \\ 22 \end{bmatrix}$$

By equality of matrices we have

$$2x - 3y = 16$$

$$x - 5y = 22$$

Solving gives $x = 2$, $y = -4$

Properties of scalar multiplications

1. If $\mathbf{A}$ and $\mathbf{B}$ are two matrices of the same order and if k is a scalar, then

$$k(\mathbf{A} + \mathbf{B}) = k\mathbf{A} + k\mathbf{B}$$

2. If k_1 and k_2 are two scalars and if A is a matrix, then

$$(k_1 + k_2)A = k_1 A + k_2 A$$

3. If k_1 and k_2 are two scalars and if A is a matrix, then

$$(k_1 k_2)A = k_1(k_2 A) = k_2(k_1 A)$$

Matrix Multiplication

The product AB of two matrices A and B is defined only when the **number of columns of A** equals the **number of rows of B** .

If A is of order $m \times n$ and B is of order $n \times p$, then AB is a matrix of order $m \times p$. Each entry of AB is obtained by multiplying the corresponding row of A with the column of B and summing the products.

Example 1.3.4: Given $A = [2 \ 3 \ 4]$ and $B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, find the product AB .

Solution: The product is a 1×1 matrix whose entry is obtained by multiplying the corresponding entries and then forming the sum.

$$\begin{aligned} AB &= [2 \ 3 \ 4] \begin{bmatrix} a \\ b \\ c \end{bmatrix} \\ &= [(2a + 3b + 4c)] \end{aligned}$$

Note that AB is a 1×1 matrix, and its only entry is $2a + 3b + 4c$.

Example 3.3.4: Given $A = [2 \ 3 \ 4]$ and $B = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$, find the product AB .

Solution: $AB = [2 \ 3 \ 4] \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix} = [10 + 18 + 28] = [56]$

Note: In order for a product of a row matrix and a column matrix to exist, the number of entries in the row matrix must be the same as the number of entries in the column matrix.

Example 1.3.5: Here is an application: Suppose you sell 3 T-shirts at \$10 each, 4 hats at \$15 each, and 1 pair of shorts at \$20. Then your total revenue is

$$\underbrace{\begin{pmatrix} 10 & 15 & 20 \end{pmatrix}}_{\text{Price}} \underbrace{\begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}}_{\text{Quantity}} = \underbrace{((10 \times 3) + (15 \times 4) + (20 \times 1))}_{\text{Revenue}} = (110)$$

Example 1.3.6: Given $A = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 3 \\ 6 & 4 \\ 7 & 5 \end{bmatrix}$, find the product AB .

Solution: We already know how to multiply a row matrix by a column matrix. To find the product AB , in this example, we will be multiplying the row matrix A to both the first and second columns of matrix B , resulting in a 1×2 matrix.

$$\begin{aligned} AB &= [2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 \quad 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5] \\ &= [56 \quad 38] \end{aligned}$$

We have just multiplied a 1×3 matrix by a matrix whose order is 3×2 . So unlike addition and subtraction, it is possible to multiply two matrices with different dimensions as long as the number of entries in the rows of the first matrix is the same as the number of entries in columns of the second matrix.

Activity 1.3.3: 1. Given the matrices E , F , G and H , below

$$E = \begin{bmatrix} 1 & 2 \\ 4 & 2 \\ 3 & 1 \end{bmatrix} \quad F = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \quad G = [4 \quad 1] \quad H = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

Find, if possible. a) GH b) FH c) EF d) FE

2.

Given the matrices R , S , and T below.

$$R = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 5 \\ 2 & 3 & 1 \end{bmatrix} \quad S = \begin{bmatrix} 0 & -1 & 2 \\ 3 & 1 & 0 \\ 4 & 2 & 1 \end{bmatrix} \quad T = \begin{bmatrix} -2 & 3 & 0 \\ -3 & 2 & 2 \\ -1 & 1 & 0 \end{bmatrix}$$

Find $2RS - 3ST$.

We summarize matrix multiplication as follows:

In order for product AB to exist, the number of columns of A , must equal the number of rows of B . If matrix A is of dimension $m \times n$ and B of dimension $n \times p$, the product AB will have the dimension $m \times p$. Let $A = (a_{ij})$ be an $m \times n$ matrix and $B = (b_{jk})$ be an $n \times p$ matrix. Then the product AB is the $m \times p$ matrix defined by $AB = (c_{ik})$, where

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \dots + a_{in}b_{nk} = \sum_{j=1}^n a_{ij}b_{jk}, \quad i = 1, 2, \dots, m \text{ and } k = 1, 2, \dots, p$$

Thus, the product $\mathbf{AB}$ is the $m \times p$ matrix, where each entry c_{ik} of $\mathbf{AB}$ is obtained by multiplying corresponding entries of the i^{th} row of $\mathbf{A}$ by those of the k^{th} column of $\mathbf{B}$ and then finding the sum of the results.

Activity 1.3.4: 1. If a matrix $\mathbf{A}$ is 3×5 and the product $\mathbf{AB}$ is 3×7 , then what is the order of $\mathbf{B}$?

Solution

Matrix multiplication dimensions

Let $\mathbf{A}$ be 3×5 . Write $\mathbf{B}$ as $r \times s$. For the product $\mathbf{AB}$ to be defined, the number of columns of $\mathbf{A}$ must equal the number of rows of $\mathbf{B}$, so $5 = r$. The product $\mathbf{AB}$ then has size $3 \times s$, and we are told $\mathbf{AB}$ is 3×7 , so $s = 7$.

Answer: $\mathbf{B}$ is 5×7 .

2. How many rows does $\mathbf{X}$ have if $\mathbf{XY}$ is a 2×6 matrix?

Solution

Matrix dimensions

Let $\mathbf{X}$ be an $m \times n$ matrix and $\mathbf{Y}$ be an $n \times p$ matrix. The product $\mathbf{XY}$ then has size $m \times p$.

Here $\mathbf{XY}$ is 2×6 , so $m=2$.

Answer: $\mathbf{X}$ has **2 rows** (i.e., $\mathbf{X}$ is $2 \times n$ for some n).

Remark: The definition refers to the product $\mathbf{AB}$, in that order, $\mathbf{A}$ is the left factor called **pre factor** and $\mathbf{B}$ is the right factor called **post factor**.

Example 1.3.7: Find the product $\mathbf{AB}$ if $\mathbf{A} = \begin{bmatrix} 1 & -4 \\ 5 & 3 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -2 & 4 & 1 & 6 \\ 2 & 7 & 3 & 8 \end{bmatrix}$

Solution: Since the number of columns of $\mathbf{A}$ is equal to the number of rows of $\mathbf{B}$, the product $\mathbf{AB} = \mathbf{C}$ is defined. Since $\mathbf{A}$ is 3×2 and $\mathbf{B}$ is 2×4 , the product $\mathbf{AB}$ will be 3×4

$$\text{Thus } \mathbf{AB} = \begin{bmatrix} -10 & -24 & -11 & -26 \\ -4 & 41 & 14 & 54 \\ 4 & 14 & 6 & 16 \end{bmatrix}$$

Observe that the product $\mathbf{BA}$ is not defined since the number of columns of $\mathbf{B}$ is not equal to the number of rows of $\mathbf{A}$. This shows that matrix multiplication is not commutative. That is, for any two matrices $\mathbf{A}$ and $\mathbf{B}$, it is usually the case that $\mathbf{AB} \neq \mathbf{BA}$ (even if both products are defined).

Example 1.3.8: Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, and $B = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$, then

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, BA = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}. \text{ Thus, } AB \neq BA.$$

Activity 1.3.4: 1. Which of the following are defined?

i) $(5 \ 2 \ 10) \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

iii) $\begin{pmatrix} a \\ b \end{pmatrix} (c \ d)$

ii) $(a \ b)(c \ d)$

iv) $(a \ b) \begin{pmatrix} a \\ b \end{pmatrix}$

2. If $A = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ 3 & 5 \end{bmatrix}$, find a and b such that $AB = BA$.

Note: 1. $AB = 0$ does not necessarily imply $A = 0$ or $B = 0$.

2. $AB = AC$ does not necessarily imply $B = C$.

Example 1.3.9: 1) Let $A = \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$, then $AB = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

2) Let $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$ then

$$AB = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix} = AC. \text{ But } B \neq C.$$

Represent a linear system as $AX = B$, where **A** is the **coefficient matrix**, **X** the column vector of variables, and **B** the constants; solving the system means finding **X** with $AX = B$.

Example 1.3.10: Verify that the system of two linear equations with two unknowns:

$$\begin{aligned} ax + by &= h \\ cx + dy &= k \end{aligned} \quad \text{can be written as } AX = B, \text{ where}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{and } B = \begin{bmatrix} h \\ k \end{bmatrix}$$

Solution: If we multiply the matrices A and X, we get

$$AX = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

If $AX = B$, then

$$\begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} = \begin{bmatrix} h \\ k \end{bmatrix}$$

If two matrices are equal, then their corresponding entries are equal.

Therefore, it follows that

$$\begin{aligned} ax + by &= h \\ cx + dy &= k \end{aligned}$$

Activity 1.3.5: Express the following system as $AX = B$.

$$\begin{aligned} 2x + 3y - 4z &= 5 \\ 3x + 4y - 5z &= 6 \\ 5x \quad \quad - 6z &= 7 \end{aligned}$$

Properties of Matrix multiplication

If A, B and C are any matrices, and if I is an identity matrix, then the following hold, whenever the dimensions of the matrices are such that the products are defined.

$$A(BC) = (AB)C \quad \text{Associative Law}$$

$$A \cdot I = I \cdot A = A \quad \text{Multiplicative Identity Law}$$

(The order of I and A
is the same)

$$A(B + C) = AB + AC \quad \text{Left Distributive Law}$$

$$(A + B)C = AC + BC \quad \text{Right Distributive Law}$$

$$A \cdot \mathbf{0} = \mathbf{0} \cdot A = \mathbf{0} \quad \text{Multiplication by Zero}$$

Remark: For real numbers, a multiplied by itself n times can be written as a^n .

Similarly, a square matrix A multiplied by itself n times can be written as A^n . Therefore, A^2 means AA , A^3 means AAA and so on.

Exercise:

$$1. \text{ If } A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 1 & -3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 5 & 6 \\ -1 & 0 & 1 \\ 2 & 1 & 2 \end{bmatrix} \text{ and } C = \begin{bmatrix} -1 & -2 & 1 \\ -1 & 2 & 3 \\ -1 & -2 & 2 \end{bmatrix}$$

Find each of the following

(i) $A + B$

ii) $2B - 3C$

iii) $A + B - C$

iv) $A - 2B + 3C$

v) $2A - C$

$$2. \text{ Let } A = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 6 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 \\ 7 & 2 \end{bmatrix}$$

Find the following: i) AB ii) BC iii) $(AB)C$ iv) $A(BC)$

$$3. \text{ If } A = \begin{bmatrix} 4 & -1 & -4 \\ 4 & 0 & -4 \\ 3 & -1 & -3 \end{bmatrix}, \text{ compute } A^2. \text{ Is it equal to } I_3, \text{ where } I_3 \text{ is the identity matrix of order 3?}$$

Transpose of a matrix

Definition 1.3.1: Let A be an $m \times n$ matrix. The transpose of A , denoted by A' or A^t , is the $n \times m$ matrix obtained from A by interchanging the rows and columns of A . Thus the first row of A is the first column of A^t , the second row of A is the second column of A^t and so on.

Example 1.3.11: If $A = \begin{pmatrix} 2 & -4 & 6 \\ 3 & 1 & 4 \end{pmatrix}$, then $A^t = \begin{pmatrix} 2 & 3 \\ -4 & 1 \\ 6 & 4 \end{pmatrix}$

Activity 1.3.6: Find a 3x3 matrix A for which $A = A^t$.

Properties: a) If A and B have the same order, $(A \pm B)^t = A^t \pm B^t$.

b) For a scalar k , $(kA)^t = kA^t$.

c) If A is $m \times n$ and B is $n \times p$, then $(AB)^t = B^t A^t$.

d) $(A^t)^t = A$

Definition 1.3.2: A square matrix A is said to be **orthogonal** if $AA^t = A^t A = I$

Example 1.3.12: $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ is orthogonal (verify)

Definition 1.3.3: A square matrix $A = (a_{ij})$ is said to be **symmetric** if $A^t = A$, or

equivalently, if $a_{ij} = a_{ji}$ for each i and j .

Example 1.3.13: $A = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & -3 \\ 5 & -3 & 6 \end{pmatrix}$ is symmetric.

Activity 1.3.7: 1. For $A = \begin{pmatrix} a & 3 & 4 & 8 \\ b & c & -3 & 9 \\ d & e & f & 10 \\ g & h & i & j \end{pmatrix}$ is to be a symmetric matrix, what numbers

should the letters a to j represent?

2. a) Does a symmetric matrix have to be square?

b) Are all square matrices symmetric?

Definition 1.3.4: A square matrix $A = (a_{ij})$ is said to be **skew symmetric** if $A^t = -A$,

or equivalently, if $a_{ij} = -a_{ji}$ for each i and j

Remark: $a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0$ or $a_{ii} = 0$. Hence elements of main diagonal of a skew-symmetric matrix are all zero.

Example 1.3.14: For $A = \begin{pmatrix} 0 & 5 & 7 \\ -5 & 0 & 3 \\ -7 & -3 & 0 \end{pmatrix}$, $A^t = \begin{pmatrix} 0 & -5 & -7 \\ 5 & 0 & -3 \\ 7 & 3 & 0 \end{pmatrix} = -A$.

So A is skew-symmetric.

Properties of symmetric and skew-symmetric matrices

1. For any square matrix A , $A + A^t$ is symmetric and $A - A^t$ is skew-symmetric
2. If A and B are two symmetric (or skew symmetric) matrices of the same order, then so is $A + B$.

Proof

i) Symmetric matrices

By definition, a matrix A is symmetric if $A^t = A$.

Similarly, B is symmetric if $B^t = B$.

Consider their sum:

$$(A+B)^t = A^t + B^t$$

Since $A^t = A$ and $B^t = B$, we get:

$$(A+B)^t = A+B$$

Therefore, $A + B$ is symmetric.

ii) Skew-symmetric matrices

By definition, a matrix A is skew-symmetric if $A^t = -A$. Similarly, $B^t = -B$.

Consider their sum:

$$(A + B)^t = A^t + B^t$$

Substituting, we get:

$$(A + B)^t = -A + (-B) = -(A + B)$$

Therefore, $A + B$ is skew-symmetric.

3. If A is symmetric or skew symmetric, then so is kA .

Proof

i) Symmetric matrix

By definition, A is symmetric if $A^t = A$.

Consider kA . Its transpose is:

$$(kA)^t = kA^t$$

Since $A^t = A$, we get:

$$(kA)^t = kA$$

Therefore, kA is symmetric.

ii) Skew-symmetric matrix

By definition, A is skew-symmetric if $A^t = -A$.

Consider kA . Its transpose is:

$$(kA)^t = kA^t$$

Since $A^t = -A$, we get:

$$(kA)^t = k(-A) = -kA$$

Therefore, kA is skew-symmetric.

4. Let A and B be symmetric matrices of the same order. The product AB is symmetric if and only if $AB = BA$.

Proof:

$$AB \text{ is symmetric} \Rightarrow (AB)^t = AB$$

$$\text{For any matrices } A, B, \text{ we know } (AB)^t = B^t A^t$$

Since $A^t = A$ and $B^t = B$, we have:

$$\begin{aligned} (AB)^t &= B^t A^t = BA \\ &\Rightarrow BA = AB \end{aligned}$$

So, AB is symmetric **if and only if** $AB = BA$.

Exercise

1. a) Form a 4 by 5 matrix, B , such that $b_{ij} = i*j$, where $*$ represents multiplication.

b) What is B^t ?

c) Is B symmetric? Why or why not?

2. Given $A = \begin{bmatrix} 3 & -1 & 0 \\ 2 & 4 & 5 \\ 1 & 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 4 & 3 \\ 5 & 1 & 7 \\ 2 & 3 & 8 \end{bmatrix}$. Verify that

i) $(A+B)^t = A^t + B^t$,

ii) $(AB)^t = B^t A^t$

iii) $(2A)^t = 2A^t$

3. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, is $A^t A$ is symmetric?

1.4 Elementary row and column operations

❖ Elementary row operations:

1. Row Replacement (Row Addition/Subtraction):

Replace one row (say R_i) by the sum of itself and a multiple of another row (say R_j).

This is abbreviated as $R_i \rightarrow k R_j + R_i$.

2. Row Interchange (Swap):

Interchange two rows of the matrix (say R_i and R_j). This is abbreviated as $R_i \leftrightarrow R_j$.

3. Row Scaling (Multiply by a Nonzero Scalar):

Multiply all entries in a row (say R_i) by a nonzero constant (scalar) k .

This is abbreviated as $R_i \rightarrow k R_i$.

❖ For elementary column operations “row” by “column” in (1), (2) and (3) above.

- Two matrices are **row equivalent** if one can be obtained from the other by a finite sequence of elementary row operations. Row operations are **reversible**:

Interchange: swapping two rows can be undone by swapping them back.

Scaling: multiplying a row by a nonzero constant k can be reversed by multiplying by $1/k$.

Replacement: if kR_i is added to R_j , the original R_j is recovered by adding $-kR_i$.

Example 1.4.1: Find the elementary row operation that transforms the first matrix in to the second, and then find the reverse row operation that transforms the second matrix in to the first.

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & -4 \\ 0 & -3 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix}$$

$$\text{Solution: } \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & -4 \\ 0 & -3 & 4 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{1}{2} R_2} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & -4 \\ 0 & -3 & 4 \end{bmatrix} \xleftarrow{2R_2 \leftarrow R_2} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix}$$

Activity 1.4.1: Find the elementary row operation that transforms the first matrix in to the second, and then find the reverse row operation that transforms the second matrix in to the first.

$$\text{a) } \begin{bmatrix} 0 & 5 & -3 \\ 1 & 5 & -2 \\ 2 & 1 & 8 \end{bmatrix}, \begin{bmatrix} 1 & 5 & -2 \\ 0 & 5 & -3 \\ 2 & 1 & 8 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 1 & -4 & 2 \\ 0 & 2 & -5 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 3 & -5 \end{bmatrix}$$

1.5 Row reduced echelon form of a matrix

In the definition that follows,

A **nonzero row (or column)** is one that contains at least one non-zero entry.

The **leading entry** of a non-zero row is its left most non-zero element.

Example

$$\begin{bmatrix} 0 & 3 & 5 \\ 2 & 4 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

- Row 1 $\rightarrow$ Leading entry = 3
- Row 2 $\rightarrow$ Leading entry = 2
- Row 3 $\rightarrow$ Leading entry = 7

Definition 1.5.1: A matrix is in **echelon form** (or **row echelon form**), if it has the following three properties:

- 1) All non-zero rows are above any rows of all zeros.
- 2) Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- 3) All entries in a column below a leading entry are zero.

If a matrix in echelon form satisfies the following additional condition then it is in **reduced echelon form** (or **row reduced echelon form**)

- 4) The leading entry in each non-zero row is 1
- 5) Each leading 1 is the only non-zero entry in its column.

Example 1.5.1: Consider the following two matrices:

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & \frac{5}{2} \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

- The first matrix is in **row echelon form (REF)**, because each leading entry is to the right of the one above it, and all zero rows (if any) are at the bottom.
- The second matrix is in **row reduced echelon form (RREF)** because each leading entry is 1, and it is the only nonzero entry in its column.

Remark: In RREF, the rules apply only to the **coefficient part** (the first three columns here). The augmented column can contain any values, because it simply shows the solution once the left side has been reduced to the identity matrix.

Definition 1.5.2: (i) A matrix which is in row echelon form is called an **echelon matrix**.

(ii) A matrix which is in row reduced echelon form is called a **reduced echelon matrix**.

Note: 1) Each matrix is row equivalent to one and only one row reduced echelon matrix.

But a matrix can be row equivalent to more than one echelon matrices.

2) If matrix A is row equivalent to an echelon matrix U, we call U an echelon form of A. If U is in reduced echelon form, we call U the reduced echelon form of A.

Activity 1.5.1: Determine which of the following matrices are in row reduced echelon form and which others are in row echelon form (but not in reduced echelon form)

a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 0 & 2 & 3 & 4 & 5 \\ 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

e) $\begin{bmatrix} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

1.6. Rank of a matrix using elementary row/column operations

Let A be an $m \times n$ matrix and U be an echelon or the reduced echelon form of A . The rank of A is denoted by $\text{Rank}(A)$ and is defined as the number of non zero rows of U .

Example 1.6.1: Find the rank of each of the matrices given in the above activity.

Solution: a) has rank 2; b) has rank 1; c, d, and e have rank 3.

Activity 1.6.1: Find the row reduced echelon form of each of the following matrices and determine the rank.

$$\text{a) } \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 3 & 5 & 7 & 9 \end{bmatrix}$$

$$\text{b) } \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{d) } \begin{bmatrix} 1 & -2 & -1 & 3 & 0 \\ -2 & 4 & 5 & -5 & 3 \\ 3 & -6 & -6 & 8 & 3 \end{bmatrix}$$